

P8X Instruction Set -- Quick Reference (rev D)

Opcodes, mnemonics and cycle counts generated live from genucode.py (the microcode source of truth) -- cannot drift from the hardware.

Operands: #imm immediate | addr 16-bit absolute | (Pn) ptr indirect | (Pn)+ post-increment | (no operand) implied. **By**=bytes, **Cy**=cycles (incl. fetch). **Flags (FI):** C carry (active-high: ADD carry-out / SUB,CMP no-borrow A>=B), Z zero, N negative (bit7), V overflow. '.' = none. Signed branches test N^V / Z. **Memory (rev D):** \$0000-3FFF ROM | \$4000-FEFF RAM | \$FF00-FFFF I/O. P0=PC, P3=stack (empty-descending). JZ/JNZ/JC are aliases of BZ/BNZ/BCP.

Op	Mnemonic	By	Cy	FI	Description
System					
\$00	NOP	1	2	-	No operation.
\$01	HLT	1	2	-	Halt clock; resume only by reset.
\$72	CLC	1	2	C	C:=0.
\$73	SEC	1	2	C	C:=1.
Interrupts (rev C)					
\$02	EI	1	2	-	Enable maskable interrupts (IE:=1).
\$03	DI	1	2	-	Disable maskable interrupts (IE:=0).
\$04	RTI	1	10	-	Return from interrupt: pop flags then PC; re-enable IE.
\$08	IRQ	1	10	-	SW interrupt: push PC+flags, vector to \$0808.
Load / store					
\$10	LDA #imm	2	2	ZN	A:=immediate.
\$11	LDB #imm	2	2	ZN	B:=immediate.
\$12	LDA addr	3	6	ZN	A:=byte at addr (absolute).
\$13	LDB addr	3	6	ZN	B:=byte at addr (absolute).
\$14	STA addr	3	6	-	byte at addr:=A (absolute).
\$15	LDA (P1)+	1	2	ZN	A:=[P1], P1++.
\$16	LDA (P2)+	1	2	ZN	A:=[P2], P2++.
\$17	LDA (P3)+	1	2	ZN	A:=[P3], P3++.
\$19	STA (P1)+	1	2	-	[P1]:=A, P1++.
\$1A	STA (P2)+	1	2	-	[P2]:=A, P2++.
\$1B	STA (P3)+	1	2	-	[P3]:=A, P3++.
\$1D	STA (P1)	1	2	-	[P1]:=A.
\$1E	STA (P2)	1	2	-	[P2]:=A.
\$1F	STA (P3)	1	2	-	[P3]:=A.
\$51	LDA (P1)	1	2	ZN	A:=[P1] (P1 kept).
\$52	LDA (P2)	1	2	ZN	A:=[P2] (P2 kept).
\$53	LDA (P3)	1	2	ZN	A:=[P3] (P3 kept).
ALU (A,B -> A)					
\$20	ADD	1	2	CZN	A:=A+B.
\$21	SUB	1	2	CZN	A:=A-B.
\$22	AND	1	2	CZN	A:=A AND B.
\$23	OR	1	2	CZN	A:=A OR B.
\$24	XOR	1	2	CZN	A:=A XOR B.
\$25	CMP	1	2	CZN	Flags from A-B; A unchanged.
\$26	INC	1	2	CZN	A:=A+1 (B unused).
\$27	DEC	1	2	CZN	A:=A-1 (B unused).
\$28	SHL	1	2	CZN	A:=A<<1, 0->bit0, out->C.
\$29	SHR	1	2	CZN	A:=A>>1, 0->bit7, out->C.
\$2A	ROL	1	2	CZN	Rotate A left through carry.
\$2B	ROR	1	2	CZN	Rotate A right through carry.
ALU with T (rev C; 2nd operand=T, B preserved)					
\$80	ADDT	1	2	CZN	A:=A+T (B preserved).
\$81	SUBT	1	2	CZN	A:=A-T (B preserved).
\$82	ANDT	1	2	CZN	A:=A AND T (B preserved).
\$83	ORT	1	2	CZN	A:=A OR T (B preserved).
\$84	XORT	1	2	CZN	A:=A XOR T (B preserved).
\$85	CMPT	1	2	CZN	Flags from A-T; A,B unchanged.
\$86	LDT #imm	2	2	-	T:=immediate.
\$87	LDT addr	3	6	-	T:=byte at addr (absolute).

Op	Mnemonic	By	Cy	FI	Description
Stack					
\$70	PHA	1	2	-	Push A onto P3 stack.
\$71	PLA	1	3	ZN	Pop A from P3 stack.
16-bit memory (rev D)					
\$74	PHW addr	3	10	-	Push 16-bit word at addr (lo then hi).
\$75	PLW addr	3	12	-	Pop 16-bit word into addr.
\$76	LPW1 addr	3	9	-	P1 := 16-bit word at addr.
\$77	LPW2 addr	3	9	-	P2 := 16-bit word at addr.
\$78	MOVW dst, src	5	13	-	16-bit mem->mem: word at src -> dst.
Control flow					
\$40	JMP addr	3	5	-	P0(PC):=addr.
\$41	JSR (P1)	1	9	-	Push return addr, P0:=P1.
\$42	RTS	1	7	-	Pop return addr from P3 into P0.
\$43	JSR addr	3	13	-	Push return addr, P0:=addr.
\$48	BZ addr	3	5	-	Branch if Z=1. (JZ alias.)
\$49	BNZ addr	3	5	-	Branch if Z=0. (JNZ alias.)
\$4A	BCP addr	3	5	-	Branch if C=1 / A>=B unsigned. (JC alias.)
\$4C	JNC addr	3	5	-	Branch if C=0 / A<B unsigned.
Signed branches (rev C; after CMP)					
\$44	BLT addr	3	5	-	Branch if signed A<B (N^V=1). After CMP.
\$45	BGE addr	3	5	-	Branch if signed A>=B (N^V=0). After CMP.
\$46	BLE addr	3	5	-	Branch if signed A<=B ((N^V)Z). After CMP.
\$47	BGT addr	3	5	-	Branch if signed A>B. After CMP.
Pointer registers					
\$31	LPL1 #imm	2	3	-	P1 low byte:=imm.
\$32	LPL2 #imm	2	3	-	P2 low byte:=imm.
\$33	LPL3 #imm	2	3	-	P3 low byte:=imm.
\$35	LPH1 #imm	2	3	-	P1 high byte:=imm.
\$36	LPH2 #imm	2	3	-	P2 high byte:=imm.
\$37	LPH3 #imm	2	3	-	P3 high byte:=imm.
\$54	INP1	1	2	-	P1:=P1+1.
\$55	INP2	1	2	-	P2:=P2+1.
\$56	INP3	1	2	-	P3:=P3+1.
\$58	DEP1	1	2	-	P1:=P1-1.
\$59	DEP2	1	2	-	P2:=P2-1.
\$5A	DEP3	1	2	-	P3:=P3-1.
\$5E	TAP1L	1	2	-	P1 low:=A.
\$5F	TAP1H	1	2	-	P1 high:=A.
\$60	TAP2L	1	2	-	P2 low:=A.
\$61	TAP2H	1	2	-	P2 high:=A.
\$62	TAP3L	1	2	-	P3 low:=A.
\$63	TAP3H	1	2	-	P3 high:=A.
\$68	TPA1L	1	2	ZN	A:=P1 low.
\$69	TPA1H	1	2	ZN	A:=P1 high.
\$6A	TPA2L	1	2	ZN	A:=P2 low.
\$6B	TPA2H	1	2	ZN	A:=P2 high.
\$6C	TPA3L	1	2	ZN	A:=P3 low.
\$6D	TPA3H	1	2	ZN	A:=P3 high.